

WASP-52b

R-0436

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_171116 · created 2026-07-19T12:55:47+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458073.65 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1663 +/- 0.0094
Transit depth (Rp/Rs)^2	2.76 +/- 0.31 [%]
Orbital Inclination (inc)	86.56 +/- 0.85
Airmass coefficient 1 (a1)	1.025 +/- 0.012
Airmass coefficient 2 (a2)	-0.017 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#3 - [387, 283]
Optimal Aperture	3.79
Optimal Annulus	9.94
Transit Duration (day)	0.0814 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1663 $\pm$ 0.0094 vs 0.1646 (deviation 0.18 σ)
Duration fit vs published	0.0814 d vs 0.0758 d (deviation 1.55 σ)
Mid-time O-C	4.9 min
Depth SNR	17.7
Residual scatter	1.15 %

Light curve

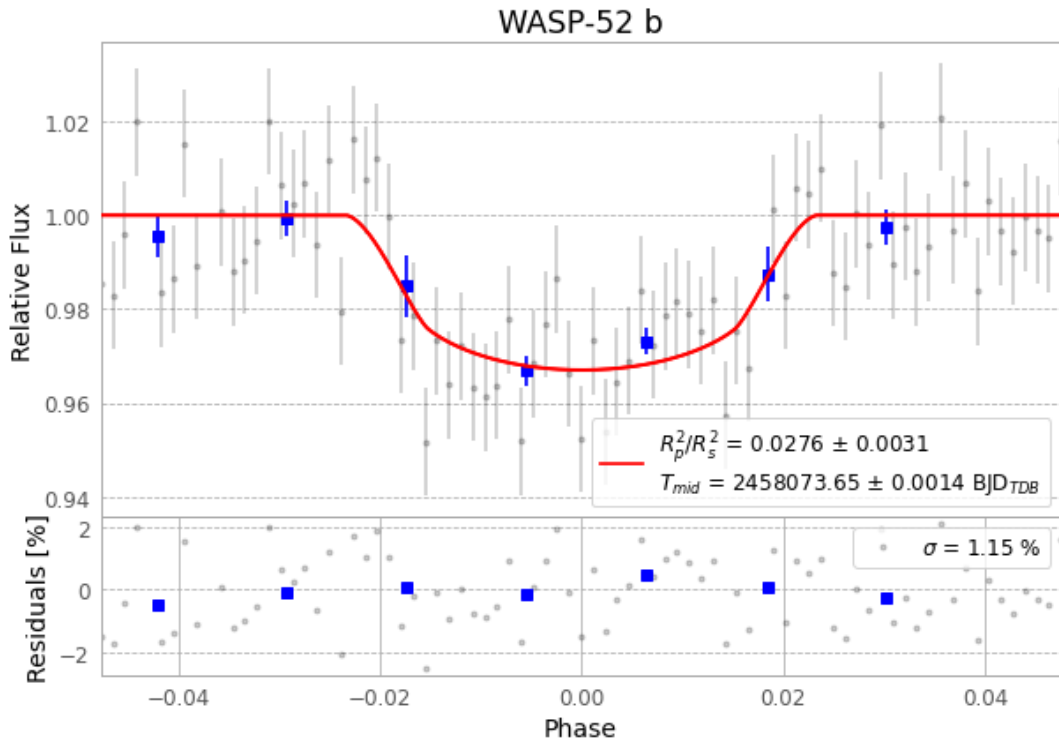


Figure 1 — FinalLightCurve_WASP-52 b_2017-11-15.png (SHA-256 931635d6fad4341d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5c6d17124c21590a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

81 FITS frames (53 MB), WASP-52171116013012.FITS → WASP-52171116053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171116.log (ed55ae96cf08...), FinalLightCurve_WASP-52 b_2017-11-15.png (931635d6fad4...), FinalLightCurve_WASP-52 b_2017-11-15.csv (db04adf15a1c...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 12:55Z.